Author response (draft, detailed)

Submission 25189

March 31, 2026

Abstract

We thank the reviewers for their careful reading and constructive feedback. This document is a **detailed** author response drafted for Overleaf: it is written in continuous prose rather than as a checklist, and each item pairs the reviewer's concern with our reply using explicit *Question.* / *Answer.* labels. The main narrative is organized around **two** primary review threads (Reviewer 1: presentation and technical exposition; Reviewer 2: novelty, scope, empirical symmetry, and scaling). Additional OpenReview accounts raised overlapping points; we answer those in later sections so that every comment receives a substantive response. A shorter digest suitable for character-limited portals is maintained separately in `answers_openreview_under5k.tex`.

Overview

We have tried to write in the spirit of strong conference rebuttals: acknowledge what is fair, correct imprecise wording, and separate what is proved from what is heuristic or empirical. Where the submission was compressed to fit the page limit, we indicate exactly what will move to the appendix or the camera-ready main text. Reproducibility hooks for new experiments are given as paths under `experiments/` when relevant.

Reviewer 1: figures, exposition, and technical foundations

(1) Figure clarity / momentum comparisons. *Question.*

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Some of the figures could be cleaner and clearer. As an example of this, in Figure 2, it is n
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Answer. We agree and will redraw Figure 2 in a clearer two-color style, with the main panel showing only matched full-rank vs low-rank curves at fixed momentum/hyperparameters. Cross-momentum sweeps will be moved to the appendix/supplementary so the main figure remains readable. We will also state in the caption that these trajectories diagnose loss-landscape/optimization effects under rank constraints; in our setup, plain SGD (no momentum) can be competitive with, or better than, momentum variants, consistent with the zero-momentum gradient-flow viewpoint of the mean-field ODE.

(2) Citation style consistency. *Question.*

```
Section 2,
you should keep the same citation style throughout the paper.
Before line 126 you cite works by mentioning the authors and then providing the reference in
```

Answer. Agreed; we will standardize the style (either always "Author (Year)" or always "[n]") throughout.

(3) "supp($W^0(C_1)$)" and frozen random features. *Question.*

```
line 139, frozen random features ensure supp(W^0(C_1)) -> this is not clear*
```

Answer. We agree the wording is unclear. What we intend is a standard *random-feature richness* assumption: with frozen first-layer features, universal approximation can be formulated via integral representations, so that the linear span of $\{\varphi_1(\langle L^0(c_1), \cdot\rangle)\}$ is dense in a suitable function class (for example $L^2(\mu_X)$ under the data distribution). This is the correct replacement for the informal "supp($W^0(C_1)$)" phrasing.

We will also *disentangle* this from assumptions on later layers, because the manuscript was easy to misread as requiring the same non-degeneracy at every depth. The camera-ready assumption block will read along the following lines: for the first layer / random feature draw ($\ell = 0$), we assume the richness condition needed for dense spans in the induced feature space; for layers $\ell > 1$, we do *not* need an analogous "full support in $\mathbb{R}^d$" statement for the trained objects. Instead, the deep part of the analysis is driven by **bounded mixing**: $|W^{(\ell)}_{c,k}| \leq K$ and hence $\|W^{(\ell)}\|_{\infty,1} \leq rK$, which controls channel aggregation and stability. Intuitively, bounded later-layer mixing interacts with first-layer richness rather than substituting for it. This matches the proof: dense random features provide approximation capacity at the input, while bounded $W^{(\ell)}$ controls the channelized Lipschitz/coupling chain.

(4) "mean-field width limit" terminology. *Question.*

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line 110, second column, what do you mean by the mean-field width limit ? Do you mean the lim
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Answer. Yes: this is a typo. By "mean-field width limit" we mean the *infinite-width limit under the mean-field parameterization* (in the neuronal-embedding sense), where empirical averages over neurons converge to expectations under a limiting neuron measure, yielding a deterministic mean-field ODE. We will reword this explicitly in the camera-ready as "infinite-width (mean-field) limit" and clarify that this refers to the mean-field scaling, *not* to NTK/lazy scaling and *not* to μP.

(5) Matrix-form rewriting suggestion. *Question.*

```
Section 3.1.
I understand that the low rank factorization appears from the stacking of W_j^{(\ell)} and h^
```

Answer. This is a good suggestion, and we agree the compact matrix view can help readers parse the architecture quickly. One can indeed rewrite the layer map as

$$\varphi\big(W^\top (Ax + c) + b\big) = \varphi(Rx + d),$$

where $R := W^\top A$ is low-rank, and the effective bias term $d := W^\top c + b$ lives in a low-rank *affine* subspace (for fixed rank) and can be interpreted as a structured "shift" at each layer.

Our reason for nevertheless keeping the channelized form in the paper is that it exposes the r *partial functions* f_k (the channel features) that are the natural state variables of the mean-field limit and the objects we visualize/track experimentally. Concretely, the second-layer pre-activation is

$$H_2(t, c_2; X, W(t)) = \sum_{k=1}^{r} W_{c_2,k}\, f_k(t; X, W(t)), \qquad f_k(t; X, W) = \mathbb{E}_{C_1}\big[w_1(t, C_1, k)\, \varphi_1(\langle W^0(C_1), X\rangle)\big],$$

as used throughout our analysis. We will add the compact matrix form in the main text as a complement, while keeping the channel decomposition as the primary analytical representation.

(6) Neuron index notation / expectations over indices. *Question.*

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Section 3.2.
Line 166, your notations with the lower and uppercase case c_1 in W^0(c_1) is not really clea
Your expectation is taken with respect to the neurons indices ? which is strange to me ? You
```

Answer. We use the neuronal-embedding formalism of Nguyen & Pham (2023): each layer is represented by measurable maps $c \mapsto w(\cdot, c)$ on (Ω_i, ρ^i), and $\mathbb{E}_{C_i}[\cdot]$ is integration under ρ^i. At finite width, neuron labels $C_i(j_i)$ are i.i.d. draws from ρ^i, so empirical averages converge to $\mathbb{E}_{C_i}[\cdot]$ under mean-field scaling. This is a continuum law-of-parameters representation (not a combinatorial "expectation over indices"); equivalently, one may use the pushforward measure on Euclidean weights. In the camera-ready, we will remove ambiguous wording (e.g., "indices become continuous") and add a short explicit paragraph with this pushforward equivalence.

(7) Write the loss and make the gradient-flow structure explicit. *Question.*

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Section 3.3.
lines 187-188, I strongly recommend writing down the expression of the loss at least and reca
```

Answer. We thank the reviewer for this suggestion and will make the statement explicit. We will define the population loss

$$\mathscr{L}(W) = \mathbb{E}_Z\big[\mathcal{L}(Y, \hat{y}(X; W))\big], \qquad d_L(Z; W) = \partial_{\hat{y}} \mathscr{L}(Y, \hat{y}(X; W)),$$

and state that the mean-field ODE is the gradient flow (with learning-rate schedules $\xi_\ell(t)$) of $\mathscr{L}$ in the mean-field parameterization. These definitions are currently spelled out in the appendix; for the camera-ready, we will also add them in the main body (with concise notation) so Section 3 is self-contained.

(8) Learning-rate schedules. *Question.*

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lines 187-188: What are the learning rate schedules?
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Answer. We will define $\xi_\ell(t) \geq 0$ as the learning-rate prefactor in the mean-field ODE (allowing time-dependent schedules such as constant, piecewise, exponential, or cosine forms). We will also state the discrete-time correspondence explicitly: with step size ϵ and iteration index n, Euler discretization gives

$$W_{n+1} = W_n - \epsilon\, \xi_\ell(\epsilon n)\, \big(\text{drift at time } \epsilon n\big),$$

and the mean-field limit is $\epsilon \to 0$ with $t = \epsilon n$ (plus the usual width scaling). In the classical mean-field parameterization, this corresponds to the familiar $1/n$ learning-rate scaling at width n.

(9) "Better than null" assumption. *Question.*

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Section 3.4.
On lines 166 - 168, you say that the ‘‘Better than null’’ assumption requires the initial loss
```

Answer. We will restate this as a non-degeneracy condition: the initial predictor should perform strictly better than the trivial constant predictor (e.g. $\hat{y} \equiv \varphi_L(0)$). In the manuscript appendix, this appears as a sufficient condition ensuring the limit point is not identically zero (the "Non-Degeneracy" assumption: $\mathscr{L}(W(0)) < \mathbb{E}[\mathcal{L}(Y, \varphi_L(0))]$).

(10) ReLU vs. the "φ_2' bounded away from zero" assumption; and "high probability in r". *Question.*

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Line 178, second column, you say that for Relu, the same holds with high probability in r. I
I don't know how easily this can be done but I would recommend laying out the assumptions cle
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Answer. We will clarify both points.

- **Meaning of r.** r is the rank (number of channels) in the low-rank bottleneck. The second-layer pre-activation is a sum over r channels:

$$H_2(c_2; x, W) = \sum_{k=1}^{r} W_{c_2,k}\, f_k(x; W).$$

- **Why ReLU is excluded in the formal global-convergence proof.** Our regularity assumption in the manuscript appendix requires $\inf_u |\varphi_2'(u)| > 0$, which excludes plain ReLU (since $\varphi'(u) = 0$ for $u \leq 0$). This is used in the step "vanishing conditional gradient implies vanishing residual".
- **What we can still say for ReLU (high-level).** The only problematic case is convergence to a degenerate stationary point where all gates are off ($H_2 \leq 0$ almost everywhere). In our architecture, this corresponds to a rare alignment event across r channels; we treat the resulting high-probability-in-r claim as intuition, not as a theorem.

To make the logical use of the nonzero-derivative condition explicit, the global-optimality proof uses the implication

$$\mathbb{E}\big[d_L(Z; W) \cdot \varphi_2'(H_2(c_2; X, W)) \mid X = x\big] = 0 \quad \Longrightarrow \quad \mathbb{E}\big[d_L(Z; W) \mid X = x\big] = 0,$$

which holds when φ_2' is bounded away from 0 on the relevant set. For ReLU, the failure mode is the "dead gate" event: $\varphi_2'(H_2) = 0$ even though $d_L \neq 0$, so the conditional moment can vanish at a non-optimal point.

In RF-LR, $H_2(c_2; x, W) = \sum_{k=1}^{r} W_{c_2,k} f_k(x; W)$ is a sum over r channels. A design intuition is that sign-structured/NMF-style initialization can bias H_2 positive on target regions and reduce dead-gate behavior (not a proof step). Under symmetric random mixing (e.g., Xavier/Gaussian draws for $W_{c_2,k}$) and non-degenerate $f_k(x)$, one-neuron "off" probability is approximately 1/2; therefore the event that all channels are simultaneously dead is heuristically exponentially unlikely in r (informally $e^{-O(r)}$), i.e., concentrated on an input subset of exponentially small measure. We keep this as intuition, not theorem (GeLU satisfies assumptions directly).

We will also add a precise definition of "mixing" and the seminorm $\|W\|_{\infty,1}$ (next item).

(11) What is "mixing"? *Question.*

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I don't know how easily this can be done but I would recommend laying out the assumptions cle
```

Answer. We will define the frozen low-rank mixing matrix $W^{(\ell)}$ as the factor that recombines the r channel features into neuron pre-activations. The only structural bound we use is entrywise boundedness (equivalently the row-sum seminorm bound):

$$\|W^{(\ell)}\|_{\infty,1} := \sup_{c_\ell} \sum_{k=1}^{r} |W^{(\ell)}_{c_\ell,k}| \leq rK,$$

as stated explicitly in the manuscript appendix (Assumption "Bounded Activations and Mixing"). In particular, $W^{(\ell)}$ can be drawn from any distribution with bounded support (or any law that satisfies the same almost-sure bound).

(12) Lipschitz bound notation: $|H_\ell(W') - H_\ell(W'')|$ and what is W^ℓ. *Question.*

```
Section 3.5 - 3.6
lines 207 - 210, second column, you have |H_\ell(W') - H_\ell(W'')|< K||W^{\ell}||_{\infty, 1}
```

Answer. We agree the notation is wrong as written in the paper, and it can indeed confuse the reader. In this part of the analysis, the trainable object is the low-rank factor denoted A (while W was used elsewhere as a global network symbol), and the prefactor is a norm of the *frozen mixing* matrix, not of the trainable tuple. In the manuscript appendix ("Bi-Lipschitz property"), for $\ell = 2$ one has:

$$\left|H_2(t, c_2; X, W') - H_2(t, c_2; X, W'')\right| \leq K \, \|W^{(2)}\|_{\infty,1} \max_{1 \leq k \leq r} \mathbb{E}_{C_1}\left[|w_1'(t, C_1, k) - w_1''(t, C_1, k)|\right].$$

We will fix the main-text inequality accordingly, clearly distinguish trainable versus frozen quantities, and harmonize notation in the camera-ready (using $W_{c_2,k}$ consistently for mixing entries).

(13) What is the solution operator F? *Question.*

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lines 218 - 219, what is the solution operator F?
```

Answer. We will define F as the Picard (integral) map associated with the mean-field ODE, whose fixed point is the unique solution trajectory. This is the standard contraction/Picard setup used in Nguyen & Pham (2023) and in our manuscript appendix well-posedness proof (where $F^{(m)}$ is the m-th Picard iterate and $F^{(m)} \to W$).

(14) Minor punctuation. *Question.*

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lines 220 - 223: ``After defining norms, the solution operator F,... and the contraction opera
```

Answer. Thank you for spotting this wording issue. We will revise the sentence in the camera-ready version to correct the punctuation and improve readability.

(15) Theorem 3.2: which depths / which widths go to infinity? *Question.*

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In the statement of Theorem 3.2., when you talk about the mean field limit, you are referring
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Answer. Thank you for this clarification request. The result is intended to hold for arbitrary depth in our RF-LR architecture, with the multilayer extension detailed in the appendix. In the camera-ready version, we will rewrite the theorem statement to specify explicitly which widths tend to infinity (all hidden-layer widths under mean-field scaling) and to separate this general statement from the depth-2 notation used for the quantitative bound.

(16) "High level proof idea" opacity. *Question.*

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lines 269 -, What you call ``High level proof idea'' is pretty opaque. I think I would remove i
```

Answer. Thank you for this suggestion. In the camera-ready version, we will either remove this high-level paragraph from the main text or rewrite it so that it mirrors the formal proof structure in the appendix (dense-span argument, stationary condition, conditional-expectation step, and loss-optimality conclusion).

(17) "upstream × local" wording. *Question.*

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lines 272 - 273, ‘‘this yields \mathbb{E}_Z[upstream\times local]’’ -> what does that mean? I u
```

Answer. We will replace this by the standard phrasing "backpropagated (upstream) signal × local derivative", and we will include the explicit definition of the backprop signal. In our notation, the channel backprop signal is

$$B_k^{(\ell)}(t; X, W) = \mathbb{E}_{C_\ell}\big[W_{C_\ell,k}^{(\ell)}\, \varphi_\ell'(H_\ell(t, C_\ell; X, W))\, w_\ell(t, C_\ell)\big].$$

(18) $\partial_2\mathcal{L}$ meaning. *Question.*

```
lines 220 - 223: What does \partial_2 \mathcal{L} mean here? Does that mean the partial deriv
```

Answer. Yes. We will define $\partial_2\mathcal{L}(y, \hat{y})$ as the partial derivative with respect to the second argument (prediction).

(19) "shifts" and why heterogeneous across neurons with frozen mixing. *Question.*

```
lines 239 - 241, second column, it is not clear to me why the frozen random features yield he
```

Answer. By "shift" we mean the additive translation behavior of intermediate-layer weights under i.i.d. initialization in the full-rank setting. In Nguyen & Pham (2023) (see Section 5.1.2, degeneracy discussion), intermediate-layer dynamics can collapse so that weights evolve essentially by a *neuron-independent translation*:

$$w_i^\infty(t, C_{i-1}, C_i) - w_i^\infty(0, C_{i-1}, C_i) = w_i^{\#}(t),$$

under constant initial biases and standard architectures. This is a precise "identical shift" phenomenon.

In our RF-LR architecture, the frozen mixing matrix L breaks this symmetry at the level of pre-activations:

$$H_2(c_2; x, W) = \sum_{k=1}^{r} W_{c_2,k} f_k(x; W),$$

so different neurons c_2 compute different linear combinations of channel partial functions. In the full-rank infinite-width limit (Nguyen & Pham (2023), Theorem 54 and Corollary 30), intermediate layers can exhibit collapse: pre-activations become neuron-independent under i.i.d. initialization, and weights receive identical additive shifts (a neuron-independent translation).

Here W enforces heterogeneity *across neurons* by construction: each neuron c_2 computes a distinct channel combination $\sum_k W_{c_2,k} f_k$. Consequently, the backprop signal aggregated through $W_{C_2,k}$ (the B_k term above) differs across neurons/channels, preventing the "all neurons move by the same increment" mode. This plays the same conceptual role as Nguyen & Pham (2023)'s bidirectional diversity achieved through correlated initializations (Theorem 38), but realized architecturally rather than initializationally.

(20) Theorem 3.3 initialization over indices; meaning of $\{n_1, n_2\} \in \mathsf{Init}$. *Question.*

```
Section 3.8
In the statement of Theorem 3.3. Here as well, it is not clear why/how you initialize over th
```

Answer. Thank you for highlighting this ambiguity. Here, $\{n_1, n_2\}$ are exactly the numbers of neurons in the first and second hidden layers in the depth-2 quantitative theorem, and Init denotes a width-indexed family of initialization laws. This matches the neuronal-embedding setup used in Nguyen & Pham (2023) (Section 4.1, Definition 1), where widths index the initialization family. We will rewrite the theorem in standard finite-width language ("take widths n_1, n_2 and initialize i.i.d.") and remove the potentially confusing "index set of Init" wording in the main text. We will also state explicitly that the detailed quantitative proof is written for depth 2, while the underlying argument extends naturally to deeper architectures.

(21) "coupling procedure" and what distance D measures. *Question.*

```
Same Theorem, what do you mean by the ‘‘coupling procedure’’?
Same, What does
 refer to here ? If I’m not wrong, you don’t introduce it. I understand it measures the devia
```

Answer. We apologize for the confusing phrase "coupling procedure" in the submission: it was an editorial artifact and not intended to refer to a separate algorithm beyond the standard Nguyen & Pham (2023) construction. What we mean is the usual **finite-width initialization + embedding** viewpoint: neurons are labeled by i.i.d. draws $C_i(j_i)$ from the limiting laws, and one compares empirical neuron averages to their mean-field expectations. There is no additional ad hoc coupling step beyond what is already implicit in the mean-field scaling.

For the distance denoted schematically by D in the informal theorem statement, we will align notation with Nguyen & Pham (2023) and with our manuscript appendix: the quantitative object is a supremum-norm-type discrepancy $\mathscr{D}_T$ between finite and mean-field trajectories on $[0, T]$, and the final bound scales as $O(1/\sqrt{n_{\min}} + \sqrt{\epsilon})$ up to the usual exponential Grönwall prefactor. When we refer to Wasserstein-type metrics in prose, we mean the transport viewpoint underlying those coupled L^1 integral assumptions, not a separate ad hoc metric introduced only for exposition.

Reviewer 2: novelty, scope, symmetry, scaling, and positioning

(1) Novelty incremental / freezing limitation / Theorem 4.1 scope. *Question.*

```
The theoretical novelty is somewhat incremental, primarily extending the framework of Nguyen

The global convergence guarantee relies critically on freezing the mixing matrices
 as random features. This architectural restriction significantly limits the theory’s applica

Theorem 4.1 provides an elegant explanation for feature learning, but it is rigorously proved
```

Answer. We agree with the spirit of this comment: the proof template is recognizably related to Nguyen & Pham (2023), and the two-point theorem is a minimal rigorous instance rather than a full theory of representation learning on natural data. Where we respectfully disagree is the framing that freezing is *only* a limitation. For our goals, freezing the mixing matrices is simultaneously a modeling choice and a technical enabler. It is what makes the deep mean-field description closed while still producing nontrivial, channel-resolved dynamics; it enforces enough heterogeneity across neurons that the degenerate "everyone receives the same effective update" modes emphasized in Nguyen & Pham (2023) are not the right mental model for the architecture we analyze.

Concretely, the non-verbatim mathematical content relative to Nguyen & Pham (2023) is not a single lemma in isolation but the *redefinition of the forward and backward objects*: the channelized map $H_\ell = \sum_{k=1}^{r} W^{(\ell)}_{\cdot,k} f^{(\ell)}_k$ propagates discrepancies through $\|W^{(\ell)}\|_{\infty,1}$ and a

$\max_{k\leq r}$ over channels (Bi-Lipschitz/coupling chain in our manuscript appendix). Frozen first-layer features also remove one delicate homotopy step used in full-rank analyses to keep first-layer support from collapsing during training, because dense-span random-feature richness can be enforced at initialization and preserved by freezing W^0.

We will revise the introduction and discussion to make the **target domain** explicit. We do not claim to subsume contemporary end-to-end training of billion-parameter models; we aim to contribute a theoretically grounded route to **low-rank / structured** training that is especially relevant when oscillatory or multiscale structure matters (scientific machine learning, PDE learning, and other settings where aggressively orthogonalized optimizers and full-rank dynamics are not the only relevant reference class). In that sense, the "restriction" is also an *assumption that matches a family of applications* where random features and factorized weights are already standard engineering tools.

For Theorem 4.1, we will keep the two-point, two-channel calculation as the fully proved anchor, and we will add a self-contained remark (already drafted below as Remark 1) explaining the *positive feedback* mechanism through $H_2 = \sum_k W_{c_2,k} f_k$ and ReLU gating at the ODE level. High-dimensional, overlapping features will be discussed honestly as primarily empirical and as ongoing work, not as something the current theorem already covers.

(2) Symmetry sensitivity and the exponential factor in the finite-width bound.
Question.

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questions : While symmetry is preserved at rank 10 , it becomes largely imperceptible at rank

Theorem 3.3 bounds the finite-width approximation error by
. Does the symmetry loss at  occur because the exponential factor  dominates  , causing diver
```

Answer. We take this concern seriously because it touches both *what the theorems mean* and *what the experiments show.*

On the Grönwall exponential and "symmetry loss." The exponential prefactor in Theorem 3.3-style bounds is the standard price of a worst-case stability analysis: in the quantitative coupling argument one obtains a constant of the schematic form $C_{\text{exp}} = e^{K_T(1+rK)}$ multiplying discretization and $1/\sqrt{n}$ errors. This is a conservative **mathematical** artifact. It should not be read as a causal explanation of a particular empirical symmetry diagnostic, and it is not something we use to claim that "rank must be tiny or the mean-field picture breaks." In practice, symmetry-like structure can emerge (or disappear) from optimization noise, batching, initializations, and the target function, largely **in parallel** with whatever worst-case coupling constant appears in a bound.

On rank 10 vs. 20 and robustness. We agree that any phenomenon that exists only for a razor-thin set of hyperparameters is not a convincing "mechanism." We also want to avoid dueling anecdotes: different targets, batch sizes, and initializations can change qualitative symmetry observations, and small-batch SGD already supplies many reasons for symmetry breaking even when the architecture admits symmetric solutions.

To make this concrete, we ran a focused reproducibility sweep after the review (fixed hidden width 1024, varying rank, multiple seeds) and measured an even-function symmetry diagnostic on a synthetic even target (sum of cosines). Table 1 reports mean±std over five seeds (42–46). The quantitative story is *not* monotonic in the way one might guess from a single pilot run: in this protocol, $r = 20$ produced *lower* mean symmetry error than $r = 10$, while test MSE was comparable within noise. This does not refute your experience—it underscores that the symmetry observable is **setup-dependent**—but it motivates us to report variance and to tone down any universal claim about "rank 10 good, rank 20 bad."

Table 1: Post-review symmetry diagnostic sweep (synthetic even target; MMNN with frozen features; width 1024). Mean and standard deviation over five seeds. $D_{\text{sym}} = \mathbb{E}[(f(x) - f(-x))^2]$ estimated on a grid of positive x; "rel." divides by mean output energy. (We will provide exact command lines and paths in supplementary material; we omit file references here for readability.)

Rank r	test MSE (mean±std)	D_{sym} (mean±std)	rel. D_{sym} (mean±std)
10	$1.10 \pm 0.26 \times 10^{-3}$	$1.89 \pm 1.73 \times 10^{-4}$	$1.57 \pm 1.42 \times 10^{-4}$
20	$1.00 \pm 0.22 \times 10^{-3}$	$4.18 \pm 0.58 \times 10^{-5}$	$3.49 \pm 0.48 \times 10^{-5}$

We will fold a shorter version of this discussion into the camera-ready (or supplement), and we will phrase symmetry observations as **hypotheses supported by particular protocols** rather than as necessary consequences of low rank.

(3) Scaling to higher intrinsic dimension and phase-transition question. *Question.*

```
If  rmust be kept strictly minimal to preserve mean-field behaviors, how does this framework
? If evaluated on a synthetic dataset with controlled
, is there a theoretical or empirical phase transition when
? It would be insightful to clarify if the network loses its theoretical structural benefits
 exceeds what the practical width
 can exponentially suppress
```

Answer. We agree this is one of the most important "beyond the theorem" questions. Our current mean-field guarantees are formulated for fixed rank r in the infinite-width limit; they are not, by themselves, a phase diagram in (d, r, N). We will state that limitation plainly in the discussion.

That said, we can still articulate a research roadmap that connects to work in progress. On the **NTK / kernel side**, understanding how rank interacts with input dimension requires controlling the induced kernel ensemble; related random-matrix analyses become delicate for low-rank structured weights, and we are actively building on recent frameworks for structured kernels (e.g. arXiv:2508.20036) to connect r scaling to high-dimensional concentration. On the **mean-field / Chizat side**, recent scaling-law and mean-field scaling discussions suggest that a "sweet spot" for faithful ODE descriptions can occur when rank grows sublinearly with width (informally, regimes such as $r \asymp \sqrt{\text{width}}$ are often discussed as practically relevant; see arXiv:2509.10167 for related scaling perspectives). Separately, in LoRA-style NTK analyses, one sometimes sees rank thresholds of the form $r \gtrsim M^{\alpha}$ (with α in the 1/4–1/2 range) discussed as regimes where kernel descriptions remain predictive; we treat such statements as *indicators*, not as theorems proved for our exact architecture.

Regarding the specific "exponential suppression" intuition: in the ReLU gating argument, the problematic configuration corresponds to a joint sign-alignment event across channels. The heuristic is not that optimization fails globally once r grows, but that the input-region measure where *all* channels are simultaneously dead can shrink exponentially in r under symmetric mixing assumptions. In higher intrinsic dimension, whether this remains effective depends on how fast r must grow to represent the target versus how concentration/gating geometry scales with d; this is exactly why we frame (d, r, N) as an empirical+theoretical phase-diagram question rather than a settled theorem.

Empirically, we have begun post-review sweeps on synthetic high-dimensional targets (using mixed sampling to avoid degenerate targets in high d) and we will report these curves without over-claiming a sharp phase transition unless the data clearly supports one. Any clean (d, r) transition is, at present, **conjectural** and belongs in "future work" unless we can prove it.

(4) Mechanism text (to add after Theorem 4.1; long-form).

Remark 1 (Mechanism behind the channel spike (beyond the two-point toy))**.** *The explanatory content is not the simplified two-point, two-channel statement itself, but the* positive feedback *it isolates.*

Fix a second-layer neuron index c_2 and recall the low-rank pre-activation

$$H_2(c_2; x, W) = \sum_{j=1}^{r} W_{c_2,j}\, f_j(x; W).$$

*For ReLU, gating enters mean-field gradients through $\varphi_2'(H_2) = \mathbf{1}\{H_2 > 0\}$. The channel-$k$ backward signal (cf. the ODE for $\partial_t w_1(\cdot, k)$) is built from expectations over C_2 of terms that couple **(i)** the mixing weight $W_{C_2,k}$, **(ii)** the gate $\mathbf{1}\{H_2(C_2; x, W) > 0\}$, and **(iii)** upstream factors (output error and later-layer weights).*

When channel k dominates*, those neurons c_2 that contribute to the channel-k drift are precisely those with the gate "on." Across c_2, the effective weights in the expectation are then **positively aligned** with $W_{c_2,k}$ and with $\mathbf{1}\{H_2(c_2) > 0\}$, producing a* **large** *contribution to the drift of the k-th channel, which in turn pushes f_k further in the same direction: a* reinforcing loop*. The ODE-level mechanism is dimension-agnostic; the two-point theorem is a minimal tractable instance.*

(5) "Limitations" wording: frozen mixing as an architectural restriction. *Question.*

```
Limitations:
Freezing the mixing matrices as random features---this architectural restriction significantl
```

Answer. We agree it would be misleading to suggest that every practical deep network is well modeled by completely frozen mixing. At the same time, we want to articulate the positive claim we are actually making: random features and factorized weights are not exotic—they are standard tools in scientific computing and structured learning—and the point of our analysis is that **global optimality-type conclusions can still be compatible with substantial low-rank structure** in a mean-field scaling, contrary to a casual intuition that "low rank only makes optimization worse." We will rewrite the limitations paragraph so it reads as a *scope statement* (what the theorems assume and what they imply within that class) rather than as an apology that could be mistaken for admitting the model is irrelevant to modern ML. We will also draw a clearer contrast with optimization stacks that deliberately orthogonalize updates (e.g. some large-scale training recipes), which target a different design point than the PDE / SciML settings we emphasize.

Further remarks for additional OpenReview reviewers

Reviewer Z9GA (technical scope and proof differences)

(1) Constant rank / one-factor-trained setting. *Question.*

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3) The low-rank model this paper consider have a constant rank w.r.t to the width that tends
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Answer. We agree the setting is stylized. We will clarify that this is the regime in which (i) deep mean-field dynamics are tractable and (ii) the i.i.d. collapse bottleneck is avoided by architectural heterogeneity. We will emphasize that this is a first step: understanding low-rank optimization mechanisms in a regime where a closed mean-field description exists.

(2) "Follows exactly the same route as Nguyen & Pham (2023)" / technical difference.
Question.

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Technically speaking, the proof of the Theorem 3.1-3.3 follows exactly the same route as in (
```

Answer. We will add an explicit "difference from Nguyen & Pham (2023)" paragraph in the camera-ready. The key non-verbatim changes are:

- the low-rank forward map $H_\ell = \sum_{k=1}^{r} W^{(\ell)}_{\cdot,k} f^{(\ell)}_k$ and the resulting stability bounds carry $\|W^{(\ell)}\|_{\infty,1}$ and $\max_{k\le r}$ over channels (see the Bi-Lipschitz lemma in the manuscript appendix);
- frozen features remove the need for a homotopy argument to maintain full support of trained first-layer weights (dense span is automatic at all t);
- frozen mixing breaks the i.i.d. collapse mode described in Nguyen & Pham (2023)'s degeneracy/collapse results by enforcing neuron-dependent channel recombination.

To make the coupling step fully transparent, the manuscript appendix also spells out an explicit identification between Nguyen & Pham (2023)'s "Δ_2^{H*}"-type drift and our low-rank channelwise quantity. For each channel k, define

$$\Gamma_k(t, z; W) := d_L(z; W(t))\, B^{(2)}_k(t; x, W(t)), \qquad z = (x, y),$$

so that $\partial_t w_1(t, c_1, k) = -\xi_1(t)\,\mathbb{E}_Z[\Gamma_k(t, Z; W)\,\varphi_1(\langle L^0(c_1), X\rangle)]$. The stability of $\Gamma_k(t, \cdot; W^*(t))$ under the convergence-to-limit-point assumption is obtained by the same triangle inequality as in Nguyen & Pham (2023) (expanded in the manuscript appendix), together with the low-rank Lipschitz control of $H_2(c_2) = \sum_{j=1}^{r} W_{c_2,j} f_j$ and the $\|W\|_{\infty,1}$ bound. This is the precise place where the low-rank channel structure enters the proof and is not verbatim from Nguyen & Pham (2023).

(2b) ReLU channel-sum control (used in low-rank coupling bounds). When bounding differences through a ReLU nonlinearity applied to a low-rank sum, we repeatedly use the elementary inequality

$$\big|\mathrm{ReLU}(\sum_{i=1}^{r} x_i) - \mathrm{ReLU}(\sum_{i=1}^{r} y_i)\big| \le \big|\sum_{i=1}^{r}(x_i - y_i)\big| \le \sum_{i=1}^{r}|x_i - y_i|.$$

This is the basic reason the low-rank forward map admits channelwise stability bounds in terms of $\|L\|_{\infty,1}$ and per-channel discrepancies.

(3) Difficulty of proving the mean-field limit when both factors are trained. *Question.*

```
As I mentioned in the Strengths and Weaknesses part, the architecture considered in this pape
 are trained. Could the authors elaborate more on the technical difficulty
```

Answer. Training both factors removes the fixed heterogeneity that breaks permutation symmetry and complicates the coupling estimates: both the forward map and the backprop signal depend on the evolving mixing, so the clean $\|L\|_{\infty,1}$-based Lipschitz chain no longer applies directly. Moreover, without frozen mixing, symmetry can re-emerge and collapse modes may reappear. We will state this obstacle explicitly and position fully trainable low-rank factors as future work.

(4)–(5) Spectral bias: universality, regimes, and heuristics (merged). *Question.*

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I wonder how universal is spectral bias discovered in Section 4 for low-rank training? For ex
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While I acknowledge the spectral bias for low-rank network could be interesting, the discussi
```

Answer. We thank the reviewers for connecting this discussion to the broader spectral-bias literature; Rahaman et al. (2019) on the spectral bias of neural networks (`arXiv:1806.08734`) is an apt reference for the standard Fourier / low-frequency-first picture in sufficiently wide, kernel-like regimes. We will revise Section 4 to separate claims by regime: in the NTK / lazy limit the bias is governed by the kernel spectrum and classical low-frequency-first behavior is expected; the empirical tilt we highlight toward comparatively higher-frequency structure is observed in a richer feature-learning setting for our low-rank RF map and is not claimed universal across all scalings (including arbitrary finite widths outside mean-field parameterizations or lazy limits). We agree Section 4 is qualitative and that the two-point analytic anchor is minimal; we will tone down claims and add CPWL/Fourier citations (Rahaman et al.; Montúfar; Raghu et al.). We will present two layers of intuition, both explicitly labeled as heuristic rather than theorem: (i) constraining weights to a low-dimensional subspace reduces the number of effectively independent hinge directions and can change boundary geometry and the anisotropy of Fourier decay; (ii) a "sum-over-paths" / polytope picture in depth-L ReLU nets, where lower effective rank loosely reduces co-activated hinge constraints and can alter the balance of region face dimensions, affecting how energy is distributed across spatial frequencies—this complementary route is work in progress and under refinement in the manuscript. We will also clarify that full-rank trained networks are discussed relative to the usual low-frequency bias narrative, while our low-rank observations are regime-dependent and empirical.

Reviewer ByWV (experiments, assumptions, typos, related work)

(1) Need more experiments. *Question.*

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Lack of additional experimental results on the performance of low-rank random feature neural
```

Answer. Agreed. We will add (space permitting) additional datasets and, crucially, variance over seeds/frozen draws to quantify robustness. We will also add a direct comparison where the full-rank baseline is trained from i.i.d. init in the same optimizer/schedule to test whether avoiding collapse yields empirical gains.

(2) "No convergence guarantee under i.i.d." / what is claimed. *Question.*

```
There is no convergence guarantee provided for the learning dynamics of low-rank random featu
```

Answer. Our main theorem is conditional: *if* the mean-field dynamics converges, the limit is a global minimizer within the parametrized class. We will highlight this more prominently and avoid implying unconditional convergence rates.

(3) What is new vs Nguyen & Pham (2023)? *Question.*

```
Derivations heavily rely on a previous work of Nguyen & Pham, 2023. It is very unclear what i
```

Answer. We will add a dedicated "What is new" paragraph: the novelty is the RF-LR architecture that (i) breaks the i.i.d. collapse symmetry via frozen mixing, (ii) yields tractable r-channel mean-field ODEs, and (iii) changes the coupling/Lipschitz chain through $\|W^{(\ell)}\|_{\infty,1}$ and channel maxima. The proof structure is inherited, but the key step and the modeling target differ.

(4) Put assumptions in the main text. *Question.*

```
I would strongly recommend mentioning the Assumptions in the main paper thoroughly with discu
```

Answer. Agreed. We will move a compact assumption block to the main text and add a short discussion of (i) what each assumption buys, and (ii) which are technical vs structural.

(5) Misc typos, momentum selection, Zhang et al. comparison, NP convergence assumptions, SGD. *Question.*

```
Typo at (040) "...is low rank all we need for global convergence ? Low-rank networks...": The
. substantially...".

At (044), it is posed that "can gradient-based training still converge to a global minimizer,
, implies that the low-rank structure affects the dynamics of the gradient-based optimization
. Note that the low-rank problem is different than the full-rank problem. Therefore, the func

For the Fig.
, how are different momentum rates selected for the corresponding level of rank constraints?

Do low-rank random feature networks differ from the multi-component multilayer neural network

Typo at (137),
 should be denoted by vector notation.

Could you clarify the convergence assumptions made by Nguyen & Pham (2023) for multilayer and

Nguyen & Pham, 2023 established the global convergence of multilayer neural networks trained
```

Answer.

- **Typos / notation.** Agreed; we will fix.
- **Global minimizer wording.** Agreed: low-rank and full-rank problems have different feasible sets; we will state explicitly that our global optimality is *within* the RF-LR parametrized class.
- **Momentum selection.** We will report the sweep protocol and only compare matched momentum settings in the main plots.
- **Zhang et al. (2023).** The architecture is identical to the multi-component networks of Zhang et al. (2023). Our novel contribution is the mean-field training analysis: we derive tractable MF ODEs for the r channels, prove global convergence under standard i.i.d. initialization, and uncover the channel-specialization mechanism. Zhang et al. do not study optimization.
- **Nguyen & Pham (2023) convergence assumptions.** We will add a short explanation: Nguyen & Pham (2023)'s convergence-to-limit-point condition is formulated as weighted coupled-L^1 vanishing under a coupling π_t (weights are products of downstream magnitudes), and they also require an $\operatorname{ess\,sup}$ decay of certain velocities to pass finite-t identities to the limit. In our manuscript appendix, the analogous assumption is written explicitly as the weighted integrals (**??**)–(**??**).
- **SGD vs ODE.** Yes: the mean-field ODE corresponds to the continuous-time limit of SGD with small step size; our quantitative theorem explicitly carries a $\sqrt{\epsilon}$ discretization term alongside the $1/\sqrt{n}$ width term (see the quantitative appendix).

Additional detail (long-form) on Nguyen & Pham (2023)'s convergence assumptions. Since the reviewer asked specifically "Could you clarify the convergence assumptions made by Nguyen & Pham (2023) for multilayer and two layer neural networks?", we will include a short boxed explanation in the camera-ready. The key point is that Nguyen & Pham (2023) do not assume mere moment convergence; they assume *weighted coupled-L^1 vanishing* under a coupling π_t tailored to the backprop/Lipschitz chain.

Two-layer mean-field networks (template). There exist limits $(\bar{w}_1, \bar{w}_2)$ and couplings π_t of the first-layer marginal with itself such that for $(C_1, C_1') \sim \pi_t$,

$$\mathbb{E}_{\pi_t}\Big[|\bar{w}_2(C_1)|\,|w_1(t, C_1') - \bar{w}_1(C_1)| + |w_2(t, C_1', 1) - \bar{w}_2(C_1)|\Big] \to 0,$$

and an additional time-regularity condition such as $\text{ess-sup}_t \left|\partial_t w_2(t, C_1, 1)\right| \to 0$ is imposed where needed to pass finite-t identities to the limit.

Three-layer networks (template). A coupling π_t of $\rho^1 \times \rho^2 \times \rho^3$ with itself is assumed so that weighted integrals of the gaps $|w_i(t, \cdot') - \bar{w}_i(\cdot)|$ vanish; the weights are products of downstream magnitudes (e.g. $(1 + |\bar{w}_3|)|\bar{w}_3||\bar{w}_2|$ multiplying the w_1 gap), reflecting backprop amplification.

Multilayer (L layers) template. In Nguyen & Pham (2023)'s multilayer theorem with correlated initialization (bidirectional diversity), the convergence-to-limit-point assumption is stated in the form

$$\mathbb{E}_{\pi_t}\Big[\big|w_i(t, C'_{i-1}, C'_i) - \bar{w}_i(C_{i-1}, C_i)\big| \prod_{j=i+1}^{L} \big|\bar{w}_j(C_{j-1}, C_j)\big|\Big] \to 0, \qquad i = 2, \ldots, L,$$

and similarly for w_1, together with an $\operatorname{ess\,sup}$ decay condition on the top-layer velocity. This is exactly the "weighted coupled" object that appears when bounding the difference between the finite-t drift and the limit drift in the global-optimality argument.

Relation to Wasserstein / W_4 (intuition). These assumptions are transport-like because they are written under couplings π_t. Under uniform moment bounds, convergence in a Wasserstein-p distance for a suitable p can imply such weighted L^1 integrals by Hölder; Nguyen & Pham (2023) use this kind of route as a sufficient condition. The definition, however, is the weighted coupled integrals themselves, not "moment convergence".

Reviewer A6kZ (guarantees, high-d mechanism, frozen-draw sensitivity)

(1) Convergence guarantees. *Question.*

```
. Convergence guarentees. Are there conditions under which convergence is guaranteed for RF-L
```

Answer. Our main guarantee is conditional on convergence of the mean-field dynamics. Unconditional convergence rates for deep RF-LR remain largely open; we will clarify this and avoid implying a general unconditional guarantee. We will also point to the sharpest partial results available and clarify which assumptions are sufficient for the global-optimality conclusion *given* convergence.

(2) High-dimensional extensions of the spike mechanism. *Question.*

```
2. High-dimension extensions. Does the spike learning mechanism extend to higher dimensions w
```

Answer. Yes in substance: the mechanism (channel dominance $\leftrightarrow$ amplified backward signal through the mixing matrix and ReLU gates) is dimension-agnostic at the ODE algebra level because it depends on the same identity $H_2 = \sum_k W_{c_2,k} f_k$ and the channelwise backprop signals. What is not currently proved in general d is a clean theorem isolating non-overlapping spikes; overlapping features complicate the rigorous argument. We will clarify this distinction and add empirical evidence in higher d.

(3) Sensitivity to the frozen feature draw. *Question.*

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3. Sensititivity of randomness of fixed feature maps. How sensitive are the convergence guara
```

Answer. The *analytical* conditions are not tied to one sampling recipe. Once mixing matrices are frozen, the arguments use an almost-sure channel-aggregation bound (schematically $\|W^{(\ell)}\|_{\infty,1} \leq rK$) and a richness/diversity assumption for the frozen first-layer features. In principle, each frozen $\mathbf{W}^{(\ell)}$ can be drawn from *any* law with finite support—equivalently, any distribution satisfying the same $\ell_{\infty,1}$ control almost surely—consistent with how "bounded mixing" enters the coupling estimates; there is no Gaussian-specific requirement. In our experiments we often use unbounded Gaussian initializations for simplicity at finite width; that is an engineering choice, while the theorems are phrased through stability/richness predicates.

Design-wise, nonnegative / sign-structured factorizations (*e.g.* NMF-style $\mathbf{W} \geq 0$) match the same boundedness template and are useful intuition for avoiding certain collapse modes; the ReLU caveat (vanishing conditional moments through dead gates vs. a true vanilla optimality implication) is separate and is where we only invoke a *high-probability-in-r* heuristic, as already clarified in the ReLU discussion tied to Reviewer VCqo (see also the technical appendix on ReLU below).

At finite width, train/test behavior can still depend on the realized frozen draw (empirical embeddings vs. mean-field expectations). We have *not* run a systematic post-review sweep isolating variance over *independent* frozen-feature draws with optimization held fixed. We will state this explicitly in the camera-ready, separate what is proved under boundedness/diversity from what is a mean-field idealization, and either add a compact multi-draw diagnostic if space permits or list it as concrete future work.

Technical appendix (kept for Overleaf brainstorming)

A. ReLU versus the non-zero derivative assumption (expanded)

Assumption "Bounded Activations and Mixing" in the manuscript appendix requires φ_2' to be bounded away from zero, i.e. $\inf_u |\varphi_2'(u)| \geq 1/K$, which excludes plain ReLU. This is used in the global-optimality implication

$$\mathbb{E}\big[d_L(Z;W) \cdot \varphi_2'(H_2(c_2;X,W)) \mid X = x\big] = 0 \quad \Longrightarrow \quad \mathbb{E}\big[d_L(Z;W) \mid X = x\big] = 0,$$

since $\varphi_2'(H_2)$ cannot vanish on the set where the residual is non-zero.

For ReLU, a "dead gradient" can occur at a non-optimal point if the gate is off:

$$\varphi_2'(H_2) = \mathbf{1}\{H_2 > 0\} = 0 \quad \text{on the relevant support},$$

so the conditional moment may vanish even when $d_L \neq 0$.

In our RF-LR architecture, the second-layer pre-activation is the low-rank channel sum

$$H_2(c_2;x,W) = \sum_{k=1}^{r} W_{c_2,k}\, f_k(x;W), \qquad f_k(x;W) = \mathbb{E}_{C_1}\big[w_1(C_1,k)\, \varphi_1(\langle L^0(C_1), x\rangle)\big].$$

Under symmetric mixing (e.g. Xavier/Gaussian draws for L) and a non-degenerate representation $(f_k(x))_{k\leq r}$ at a fixed x, the event $\{H_2(c_2;x,W) \leq 0\}$ is heuristically a half-space event for each independent channel contribution, so the probability that gates shut off "everywhere" in a way that blocks gradient propagation is heuristically exponentially small in r (informally $e^{-O(r)}$). We keep this only as intuition and avoid presenting it as a theorem; smooth activations (e.g. GeLU) satisfy the assumption directly.

B. "Shifts" / degeneracy under i.i.d. init versus architectural heterogeneity

In Nguyen & Pham (2023), i.i.d. initialization can induce degeneracy for deep networks: intermediate-layer pre-activations become neuron-independent in the infinite-width limit, and intermediate-layer weights can evolve essentially by a neuron-independent translation. Concretely, Nguyen & Pham (2023) show that under constant initial biases, at intermediate layers one obtains concentration around a neuron-independent limit $H_i^*(t,X,B_i)$, and in standard setups weights can satisfy a translation property of the form

$$w_i^\infty(t,C_{i-1},C_i) - w_i^\infty(0,C_{i-1},C_i) = w_i^\#(t),$$

so all neurons receive the same additive "shift".

In our RF-LR setting, frozen mixing breaks this symmetry at the level of the forward map:

$$H_2(c_2; x, W) = \sum_{k=1}^{r} W_{c_2,k}\, f_k(x; W),$$

so neurons c_2 compute distinct channel combinations and thus induce heterogeneous effective coefficients in the backpropagated signals

$$B_k^{(2)}(t; X, W) = \mathbb{E}_{C_2}\big[W_{C_2,k}\, \varphi_2'(H_2(t, C_2; X, W))\, w_2(t, C_2)\big].$$

This is the architectural analogue of the "diversity" mechanisms used in Nguyen & Pham (2023) to avoid the i.i.d. bottleneck.

C. Coupling step and low-rank substitute (expanded bookkeeping)

The convergence-to-limit-point assumption is written as weighted coupled-L^1 vanishing under a coupling π_t, in our two-layer RF-LR case:

$$\int (1 + |\bar{w}_2(c_2)|)|\bar{w}_2(c_2)| \max_{1\le k\le r} |\bar{w}_1(c_1, k)|\, |w_1^*(t, c_1', k) - \bar{w}_1(c_1, k)|\, d\pi_t(c_1, c_2, c_1', c_2') \to 0,$$
$$\int (1 + |\bar{w}_2(c_2)|)|\bar{w}_2(c_2)|\, |w_2^*(t, c_2') - \bar{w}_2(c_2)|\, d\pi_t(c_1, c_2, c_1', c_2') \to 0,$$

which are exactly the displayed assumptions (**??**)–(**??**) in the manuscript appendix.

To connect these integrals to the drift terms in the global-optimality argument, the manuscript appendix introduces the channelwise quantity

$$\Gamma_k(t, z; W) := d_L(z; W(t))\, B_k^{(2)}(t; x, W(t)), \qquad z = (x, y),$$

and uses the pointwise decomposition

$$|\Gamma_k(t, z; W) - \Gamma_k(z; \bar{W})| \le |d_L(z; W) - d_L(z; \bar{W})|\, |B_k^{(2)}(t; x, W)| + |d_L(z; \bar{W})|\, |B_k^{(2)}(t; x, W) - B_k^{(2)}(x; \bar{W})|.$$

The low-rank channel structure enters when bounding the difference of $B_k^{(2)}$: the argument of φ_2' is the channel sum $H_2 = \sum_j W_{\cdot,j} f_j$, so differences propagate through $\|W\|_{\infty,1}$ and channelwise discrepancies.

For ReLU on the *forward* map, we also use the elementary channel-sum bound

$$\big|\mathrm{ReLU}(\sum_{i=1}^{r} x_i) - \mathrm{ReLU}(\sum_{i=1}^{r} y_i)\big| \le \sum_{i=1}^{r} |x_i - y_i|.$$

D. Nguyen & Pham (2023) convergence assumptions and relation to Wasserstein (long-form)

Nguyen & Pham (2023) formulate convergence to a limit point via *weighted coupled-L^1* integrals under a coupling π_t, where the weights are products of downstream magnitudes (backprop amplification). For multilayer networks (with their notation), a typical template is

$$\mathbb{E}_{\pi_t}\Big[\big|w_i(t, C_{i-1}', C_i') - \bar{w}_i(C_{i-1}, C_i)\big| \prod_{j=i+1}^{L} \big|\bar{w}_j(C_{j-1}, C_j)\big|\Big] \to 0,$$

together with an ess sup decay condition on a top-layer velocity that is used to pass finite-t stationarity identities to the limit.

These assumptions are transport-like because they are written under couplings π_t; convergence in a Wasserstein-p distance for an appropriate p, together with uniform moment bounds, can imply such weighted L^1 integrals by Hölder. The definition, however, is the weighted coupled integrals themselves, not mere "moment convergence".